6 **(a)** Define electric potential at a point.

....

....

.... [2]

(b) An isolated conducting sphere of radius R is in a vacuum. The sphere has a charge of $+Q$ distributed around its surface, as illustrated in Figure 6.1.

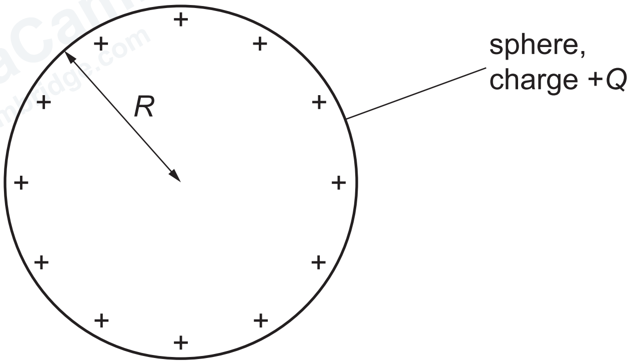

Figure 6.1

(i) State an expression, in terms of R and Q, for the magnitude V_0 of the electric potential at the surface of the sphere. Identify any other symbols that you use.

V_0 = ... [1]

(ii) On Figure 6.2, sketch the variation of the electric potential V at distance x from the centre of the sphere between $x = R$ and $x = 3R$.

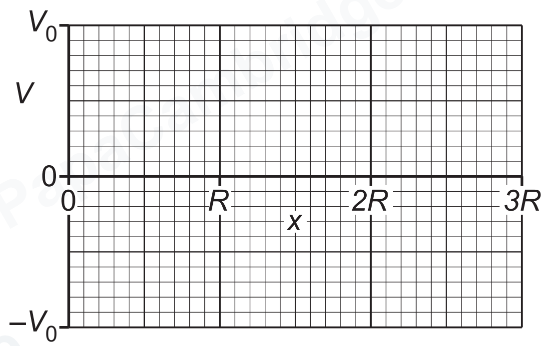

Figure 6.2

[3]

(c) The dome of an electrostatic generator, called a Van de Graaff generator, may be considered to be a conducting sphere of radius 11.9 cm.

The dome is charged so that the electric potential at its surface is -4.80×10^4 V.

(i) Calculate the charge on the surface of the dome.

charge = ... C [2]

(ii) Explain why the dome acts as a capacitor.

....

.... [1]

(iii) Calculate the capacitance of the dome. Give a unit with your answer.

capacitance = ... unit .... [2]

(iv) Determine the energy stored by the dome.

energy = ... J [2]

[Total: 13]